

Pavan Kumar

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EDUCATION

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR

GPA : 8.45

Bachelor of Engineering

2022 - 2026

Major in Computer Science

Coursework: Data Structures, Algorithms, Software Engineering, Machine Learning, Gen AI, NLP

EXPERIENCE

AI-POWERED ENERGY MANAGEMENT AND FORECASTING PLATFORM(2nd Place, LUMINOUS TECHNO X, 40,000+ teams)

- Engineered an AI-driven platform for energy forecasting, anomaly detection, load balancing, and real-time monitoring.
- Implemented LSTM-based predictive models for tariff and consumption forecasting, enhancing energy efficiency.
- Delivered a scalable full-stack solution with **Vite + React, PostgreSQL (Neon), Grafana, and Vercel.**

PROJECTS

AI-Powered Quiz Generation Platform

May 2025

- Built an intelligent quiz generation system with a **Flask (Python)** and responsive **HTML/CSS/JavaScript**.
- Implemented document parsing (PDF, DOCX, TXT) and token-based chunking using **PyMuPDF, python-docx,** and **tiktoken**.
- Developed **RESTful APIs** for document upload and quiz generation, with planned **Phi-3-mini** and **FAISS** integration for RAG-based AI quizzes.

Vision Tracking Model with Deep Learning

Dec 2024

- Designed a **deep learning-based eye-tracking system** for **driver alertness assessment**, detecting drowsiness via pupil tracking and gaze estimation.
- Applied **CNN models (Conv2D, MaxPooling, Dense layers)** using **Python, PyTorch, and OpenCV** for pupil coordinate prediction.
- Trained on the **LPW dataset**, achieving optimized accuracy for **real-time fatigue detection**.

ACTIVITIES

- Explore **personal development projects** to experiment with new technologies and apply theoretical knowledge in practice.
- Completed **300+ Data Structures and Algorithms problems** on **LeetCode** and **GeeksforGeeks**.
- Served as **Science & Technology Secretary** at IIT Raichur, leading student initiatives and coordinating technical activities.

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript, HTML, CSS, PostgreSQL

Web & Software Development: React.js, Node.js, Express.js, Next.js, REST APIs, Object-Oriented Programming (OOP), Git

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Scikit-learn, XGBoost, PyTorch, LangChain, Large Language Models (LLMs), Hugging Face

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Power BI